

Exploratory Data Analysis (EDA) & Predictive Model Plan

1. Introduction

The purpose of this report is to conduct an exploratory data analysis (EDA) on Geldium’s delinquency prediction dataset and propose a predictive modeling approach. The analysis identifies missing values, anomalies, and early delinquency risk indicators to support Tata iQ’s analytics team in improving intervention strategies and refining predictive models.

2. Dataset Overview

The dataset contains **500** customer records and 19 variables related to customer demographics, financial behavior, and payment history. Numerical and categorical variables are included to support delinquency prediction analysis.

Key dataset attributes:

- Number of records: 500
- Data types: Numerical, categorical, and binary variables
- Duplicate records identified: 0 duplicates detected

Variable	Description	Type
Income	Annual customer income	Numerical
Credit_Score	Customer credit score	Numerical
Credit_Utilization	Percentage of available credit used	Numerical
Missed_Payments	Count of missed payments	Numerical
Delinquent_Account	Delinquency indicator (0/1)	Binary
Employment_Status	Employment category	Categorical
Month_1 - Month_6	Monthly payment history	Categorical

3. Missing Data Analysis

The dataset contains missing values in several financial variables that could impact predictive model accuracy if not addressed properly. **Key missing data findings:**

- Income: 39 missing values
- Loan_Balance: 29 missing values
- Credit_Score: 2 missing values

Variable	Treatment Method	Justification
Income	Median Imputation	Reduces sensitivity to outliers
Loan_Balance	Median Imputation	Handles skewed balance distributions
Credit_Score	Mean Imputation	Very low missing rate

4. Key Findings and Risk Indicators

Key findings:

- Customers with higher credit utilization showed slightly higher delinquency rates.
- High debt-to-income ratios were associated with increased repayment risk.
- Repeated “Late” and “Missed” payment patterns across Month_1 to Month_6 were strong behavioral indicators.
- Shorter account tenure appeared associated with slightly higher risk.

Unexpected anomalies:

- Some customers with relatively high income levels still showed delinquency behavior.
- Credit utilization values above 100% may indicate over-limit borrowing or data inconsistencies requiring further investigation.

5. AI & GenAI Usage

Generative AI tools were used to summarize the dataset, identify anomalies, suggest imputation strategies, and highlight potential delinquency indicators. **Example prompts used:**

- “Summarize key patterns in the dataset and identify anomalies.”
- “Suggest an imputation strategy for missing income values based on industry best practices.”
- “Identify the top variables most likely to predict delinquency.”

6. Conclusion & Next Steps

The dataset is suitable for predictive modeling after preprocessing and missing value treatment. Behavioral variables such as payment history, credit utilization, and debt-to-income ratio emerged as the strongest indicators of delinquency risk. Before model development, additional preprocessing steps such as encoding categorical variables, scaling financial features, and handling class imbalance are recommended. **Recommended next steps:**

1. Complete data preprocessing and feature engineering.
2. Standardize payment history variables.
3. Investigate outlier credit utilization values.
4. Train predictive models using logistic regression, decision trees, or random forests.

Predictive Model Plan – Student Template

1. Model Logic (Generated with GenAI)

The proposed predictive model uses a supervised classification approach to predict whether a customer is likely to become delinquent. The process includes:

1. Clean and preprocess the dataset.
2. Handle missing values using imputation.
3. Encode categorical variables such as payment history and employment status.
4. Split the dataset into training and testing sets.
5. Train a logistic regression or random forest model using customer financial and behavioral variables.
6. Predict delinquency probability for each customer.
7. Flag high-risk customers for early intervention.

2. Justification for Model Choice

Logistic regression was selected as the primary baseline model because it is transparent, interpretable, and widely used in financial risk prediction. It allows analysts to understand how variables such as credit utilization and payment history influence delinquency probability. Random forest models may also be explored because they capture non-linear relationships and interactions between variables while improving predictive accuracy. These models are suitable for Geldium’s business needs because they support explainable decision-making and operational scalability.

3. Evaluation Strategy

The model performance will be evaluated using the following metrics:

- Accuracy: Measures overall prediction correctness.

- Precision: Measures how many predicted delinquent customers were truly delinquent.

- Recall: Measures the model's ability to correctly identify delinquent customers.
- F1 Score: Balances precision and recall for imbalanced datasets.
- AUC-ROC: Measures how well the model distinguishes between delinquent and non-delinquent customers.

Bias detection will include reviewing model predictions across demographic and employment groups to ensure fairness. Ethical considerations include avoiding discriminatory outcomes, maintaining customer privacy, and ensuring transparency in financial decision-making.